

Confidential Instructions:

Candidates are advised to spend no more than:

- **60 minutes** on **Question 1**.
- **50 minutes** on **Question 2**.
- **35 minutes** on **Question 3**.

Question 1

- i. 15 cm³ of 0.1% 2,6-dichlorophenol indophenol (DCPIP) solution in a corked, labelled container. Dissolve powder in distilled water.
- ii. 5 cm³ of each of four standard solutions of ascorbic acid prepared as follows:

Dissolve 400 mg of ascorbic acid powder in 100 cm³ distilled water. Add 15 drops of BDH Universal Indicator solution and adjust the pH to between 7 and 8 by adding 5% sodium hydroxide solution drop by drop. This is **4.0mgcm⁻³** ascorbic acid solution. Take 50 cm³ of this solution and to it, add 50cm³ of distilled water (the **2.0mgcm⁻³** ascorbic acid solution). Repeat this procedure to obtain **1.0mgcm⁻³** and **0.5mgcm⁻³** ascorbic acid. Any variations in the colours of the solution can be ignored, but the pH of the most dilute solution should be checked with indicator paper. If it is found to be below 7 it should be adjusted to pH 7-8 by adding further drops of 5% sodium hydroxide solution.

The solution should be dispensed to students in a corked, **labelled** tubes from which 1cm³ volumes can be withdrawn.

- iii. 5 cm³ of fresh orange juice. Dispense to students in corked tube labelled orange juice.
- iv. Approximately 10cm³ of **4% glucose solution**; 100cm³ of **distilled water**, and 50cm³ of **Benedict's solution** in appropriately labelled, corked containers.
- v. Three 2 cm³ syringe, and two 5 cm³ syringe; a glass rod; 4 small beakers; 12 test tubes
- vi. Test tube rack; Bunsen burner; test tube holder; 250cm³ beaker; marker; tripod

Question 2

- i. A 10 cm length of deeply pigmented purple onion. The plant material should not be allowed to dry out
- ii. About 5cm³ of each of the following in stoppered container labelled appropriately: distilled water, 1 moldm⁻³ sucrose solution and 1 moldm⁻³ [O,T] potassium nitrate solution
- iii. 2 clean dry microscope slides and cover slips; 4 droppers; circle of filter paper; pair of fine forceps; fine scissors

Apparatus List

Candidates will require:

Question 1

1	DCPIP	13	1 dropper
2	Ascorbic acid (AA) labelled 0.5 mgcm ⁻³ , 1.0 mgcm ⁻³ , 2.0 mgcm ⁻³ and 4.0 mgcm ⁻³	14	4 50-ml beakers
		15	1 250ml beaker
		16	1 500ml beaker
		17	1 stopwatch
		18	1 wire gauze
3	4% glucose	19	1 Bunsen burner
4	Distilled water	20	1 tripod stand
5	Benedict's solution	21	1 lighter
6	Orange juice	22	1 wash bottle labelled as distilled water
7	12 Test tubes		
8	1 test tube rack	23	1 pair of goggles
9	1 test tube holder	24	1 permanent marker
10	3 2-ml syringes		
11	2 5-ml syringes		
12	1 Glass rod		

Question 2

1	Red onions
2	1M Sucrose
3	1M Potassium nitrate (oxidising, toxic)
4	1 Petri Dish
5	2 microscopic glass slide
6	2 cover slips
7	4 droppers
8	2 filter papers
9	1 pair of scissors
10	1 forceps
11	1 mounted needle